

Storing Computer Algebra Data

- It's common to have multiple perspectives on an object in mathematics.
- While storing mathematical data a choice of perspective must be made.
- Such a choice might not be describable in an email.

Say we want to store:

$$p = 2y^3z^4 + (a + 3)z^2 + 5ay + 1$$



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Say we want to store:

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- Some technicalities with the coefficients.
- Is y considered a coefficient of z ?
- What is a ?
- How can we guarantee the objects behave as expected on load?

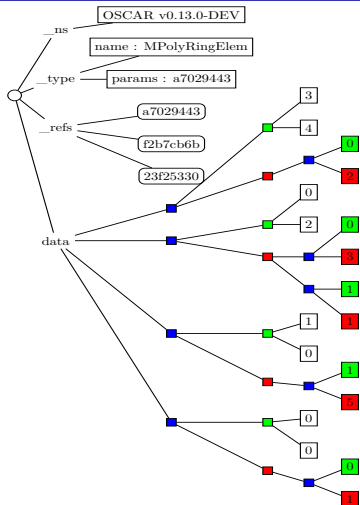


The mrdi File Format

- JSON file format.
- Similar to polymake format but generalizes to include algebraic data.
- Namespaces for semantic separation.
- References stored with UUIDs.



Tree Structure



$$2y^3z^4$$

$$(a + 3)z^2$$

5ay

1



Example File Serialized with OSCAR

```
{
  "_ns": { "Oscar": [ "https://github.com/oscar-system/Oscar.jl", "1.0.0" ] },
  "_type": {
    "name": "MPolyRingElem",
    "params": "869a359a-43d3-43f4-9821-0af9346be019"
  },
  "data": [[["3", "4"], [{"0", "2"}]],
    [{"0", "2"}, [{"0", "3"}, ["1", "1"]]],
    [{"1", "0"}, [{"1", "5"}]],
    [{"0", "0"}, [{"0", "1"}]],
  "_refs": {
    "152ac7bd-e85a-4b36-acc2-743ade2cad4f": {
      "data": { "base_ring": { "data": "7", "_type": "FqField"},
        "symbols": ["x"] },
      "_type": "PolyRing"
    },
    "869a359a-43d3-43f4-9821-0af9346be019": {
      "data": {
        "base_ring": "a8309b96-caec-443c-bedb-e23bb0634c14",
        "symbols": [ "y", "z" ]
      },
      "_type": "MPolyRing" },
    "a8309b96-caec-443c-bedb-e23bb0634c14": {
      "data": {
        "def_pol": {
          "data": [{"0", "1"}, ["2", "1"]],
          "_type": {
            "name": "PolyRingElem",
            "params": "152ac7bd-e85a-4b36-acc2-743ade2cad4f"
          }
        },
        "_type": "FqField"
      }
    }
  }
}
```



Namespaces

- We make no attempt to formalise the language of Mathematical Research Data.
- We rely on namespaces to pin the semantics to a specific software.
- Open source software provides example implementations for saving and loading.
- Freedom to upgrade encoding.
- Similar with PDF.



Serialized Toric Divisor

```
{
  "_ns": { ...
  "_type": { "name": "ToricDivisorClass", "params": "b66b610c-f21b-4b3d-8b1f-267e8e65ec7" },
  "data": [ "5", "0", "0" ],
  "_refs": {
    "b66b610c-f21b-4b3d-8b1f-267e8e65ec7a": {
      "_type": "NormalToricVariety",
      "data": {
        "_ns": { "polymake": [ "https://polymake.org", "4.12" ] },
        "_type": "fulton::NormalToricVariety",
        "FAN_AMBIENT_DIM": 2,
        "PROJECTIVE": true,
        "MAXIMAL_CONES": [ ... ],
        "PSEUDO_REGULAR": true,
        "COMPLETE": true,
        "DIVISOR": [ ... ],
        "RAYS": [ [ "1", "0" ], [ "0", "1" ], [ "-1", "-1" ] ],
        ...
      }
    }
  }
}
```



Data Collections

- Johnson Solids.
- Non-general type surfaces in $\mathbb{P}^4$.
- F-Theory models.
- Small Phylogenetic Trees. (Luis David Garcia-Puente, Jacob Porter)



Thank You!

